2510514021

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Parameter Personalisasi

$N = 21,\ a = 2,\ b = 1,\ k = 4,\ \theta^\circ = 65^\circ,\ \alpha = 0{,}002$

a) $A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} = \begin{bmatrix} 2+1 & 1 & 2 \\ 1 & 2+2 & 1 \\ 1 & 1 & 2+3 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 1 & 5 \end{bmatrix}$

$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

$M_{11} = \begin{vmatrix} 4 & 1 \\ 1 & 5 \end{vmatrix} = (4)(5) - (1)(1) = 20 - 1 = 19$

$C_{11} = (-1)^{1+1}(M_{11}) = (-1)^2(19) = 19$

$M_{12} = \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} = (1)(5) - (1)(1) = 5 - 1 = 4$

$C_{12} = (-1)^{1+2}(M_{12}) = (-1)^3(4) = (-4)$

$M_{13} = \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix} = (1)(1) - (4)(1) = 1 - 4 = (-3)$

$C_{13} = (-1)^{1+3}(M_{13}) = (-1)^4(-3) = (-3)$

$\det A = 3(19) + 1(-4) + 2(-3) = 57 + (-4) + (-6) = 47$

b) $M_{11} = \begin{vmatrix} 4 & 1 \\ 1 & 5 \end{vmatrix} = (4)(5) - (1)(1) = 20 - 1 = 19$

$M_{12} = \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} = (1)(5) - (1)(1) = 5 - 1 = 4$

$M_{13} = \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix} = (1)(1) - (4)(1) = 1 - 4 = (-3)$

$M_{21} = \begin{vmatrix} 1 & 2 \\ 1 & 5 \end{vmatrix} = (1)(5) - (2)(1) = 5 - 2 = 3$

$M_{22} = \begin{vmatrix} 3 & 2 \\ 1 & 5 \end{vmatrix} = (3)(5) - (2)(1) = 15 - 2 = 13$

$M_{23} = \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} = (3)(1) - (1)(1) = 3 - 1 = 2$

$M_{31} = \begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix} = (1)(1) - (2)(4) = 1 - 8 = -7$

$M_{32} = \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = (3)(1) - (2)(1) = 3 - 2 = 1$

$M_{33} = \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} = (3)(4) - (1)(1) = 12 - 1 = 11$

$M_{ij} = \begin{vmatrix} 19 & 4 & -3 \\ 3 & 13 & 2 \\ -7 & 1 & 11 \end{vmatrix} \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} = \begin{vmatrix} 19 & -4 & -3 \\ -3 & 13 & -2 \\ -7 & -1 & 11 \end{vmatrix}$

c) Adj(A) = kof A → baris jadi kolom

$\mathrm{Adj}(A) = \begin{bmatrix} 19 & -3 & -7 \\ -4 & 13 & -1 \\ -3 & -2 & 11 \end{bmatrix}$

$A^{-1} = \dfrac{1}{\det(A)}\ \mathrm{Adjoin}(A)$

$= \dfrac{1}{47} \cancel{\begin{vmatrix} 19 & -4 & -3 \\ -3 & 13 & -2 \end{vmatrix}} \begin{vmatrix} 19 & -3 & -7 \\ -4 & 13 & -1 \\ -3 & -2 & 11 \end{vmatrix} = \begin{vmatrix} 19/47 & -3/47 & -7/47 \\ -4/47 & 13/47 & -1/47 \\ -3/47 & -2/47 & 11/47 \end{vmatrix}$

2) $P = \begin{vmatrix} 120 & (50+N) & 200 \\ 80 & 2+2 & (10+N) \\ (30+N) & 90 & 170 \end{vmatrix} \cdot \begin{vmatrix} 120 & 50+21 & 200 \\ 80 & 2+2 & 10+21 \\ 30+21 & 90 & 170 \end{vmatrix}$

$= \begin{vmatrix} 120 & 71 & 200 \\ 80 & 4 & 31 \\ 51 & 90 & 170 \end{vmatrix}$

1. $E = A.P \rightarrow \begin{vmatrix} 3 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 1 & 5 \end{vmatrix} \begin{vmatrix} 120 & 71 & 200 \\ 80 & 4 & 31 \\ 51 & 90 & 170 \end{vmatrix}$

$E11 = 3.120 + 1.80 + 2.51 = 360 + 80 + 102 = 542$
$E12 = 3.71 + 1.4 + 2.90 = 213 + 4 + 180 = 397$
$E13 = 3.200 + 1.31 + 2.170 = 600 + 31 + 340 = 971$
$E21 = 1.120 + 4.80 + 1.51 = 120 + 320 + 51 = 491$
$E22 = 1.71 + 4.4 + 1.90 = 71 + 16 + 90 = 177$
$E23 = 1.200 + 4.31 + 1.170 = 200 + 124 + 170 = 494$
$E31 = 1.120 + 1.80 + 5.51 = 120 + 80 + 255 = 455$
$E32 = 1.71 + 1.4 + 5.90 = 71 + 4 + 450 = 525$
$E33 = 1.200 + 1.31 + 5.170 = 200 + 31 + 850 = 1081$

$E = \begin{vmatrix} 542 & 397 & 971 \\ 491 & 177 & 494 \\ 455 & 525 & 1081 \end{vmatrix}$

Pembuktian $P = P'$

$P' = \frac{1}{47} \begin{vmatrix} 19 & -3 & -7 \\ -4 & 13 & -1 \\ -3 & -2 & 11 \end{vmatrix} \begin{vmatrix} 542 & 397 & 971 \\ 491 & 177 & 494 \\ 455 & 525 & 1081 \end{vmatrix}$

Test Diagonalnya

$P11 = 19 \times 542 + (-3 \times 491) + (-7 \times 455) = 10.298 + (-1.473) + (-3.185)$
$= 5.640 \div 47 = 120$
$P12 = (-4 \times 397) + 13 \times 177 + (-1 \times 525) = (-1.588) + 2301 - 525$
$= 188 \div 47 = 4$
$P13 = (-3 \times 971) + (-2 \cdot 494) + (11 \cdot 1081) = (-2913) + (-988) + 11.891$
$= 7990 \div 47 = 170$

$P = \begin{vmatrix} 120 & & \\ & 4 & \\ & & 170 \end{vmatrix} \quad P' = \begin{vmatrix} 120 & & \\ & 4 & \\ & & 170 \end{vmatrix}$

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Diskusi : Apa yang terjadi jika det(A) = 0?
: Jika det(A) = 0, maka matriks tersebut bersifat singular dan tidak memiliki invers. File citra tidak bisa dikembalikan ke bentuk awal melalui inverse transformasi, tetapi bisa diolah dengan operasi digital lainnya